Analysis on local news outlet in comparison to global news outlet in word usage.

Project Proposal

1. Aims, objectives and background

1.1 Introduction

With the invention of the internet, access to the vast information highway has transformed the media landscape, compelling it to adapt to the new digital world. Even before the advent of the internet, media played a crucial role in shaping public perception and understanding of events. However, the internet has significantly amplified this influence by providing real-time access to global happenings. Despite covering the same topics or news reports, the way news is reported can vary significantly between local and global news outlets. This project aims to analyze and compare word usage in local versus global news outlets to uncover patterns, differences, and similarities in how news is communicated.

This project will employ several methods to analyze a diverse range of news articles from both local and global news outlets. We will scrape text from articles across different news outlets, both local and global, and examine word frequency, as well as the use of specific jargon or terminology. The data will be sourced from Singaporean and United Kingdom news outlets, where British English is predominantly used. By selecting three well-known news outlets from each country, we aim to provide a more accurate representation of word usage among the populace.

<h3 style="font-weight: bold"h3>1.2 Aims and objectives</h3> Within this project, I would like to explore the following:

- Data Collection and Preprocessing
  - To gather dataset of articles from selected local and global news outlets to see the differences between the two
  - To clean and preprocess the text data to ensure accuracy and consistency in the analysis

- Frequency Analysis
  - To perform word frequency analysis on the datasets to identify the most commonly used words in local and global news
  - To create comparative visualizations, such as word clouds and bar charts, to highlight key differences in word usage.

1.3 Data

1.3.1 Data Requirements

Three publications were chosen from each country, allowing a reasonable sample size from each country's publication. Choosing 3 types of article topics from each news publication (alternative are chosen if none are found for the particular news outlet). This allow for similar topics while having enough sample of article from each country to explore the similarity and differences. About 130

articles are chosen from each news publication which would be allow for about 300 to 400 articles for each country.

1.3.2 Choice of news outlet

In order to represent each country's publication outlet accurately, news outlet are chosen by the popularity of said news outlet within the country:

- Local
    - The Straits Times: Leading the ranks as the top Singaporean web publisher, The Straits Times boasts an impressive 3.2 million engagements.[1]
    - Today Online: Positioned 8th on the list, Today Online has garnered a notable 30 thousand engagements [1]
    - Channel News Asia: Holding the 3rd spot, Channel News Asia achieves a substantial 1.6 million engagements [1]

- Global
    - Daily Mail: Ranked 5th in influence score, the Daily Mail stands among the top media outlets [2]
    - The Guardian: Holding the 2nd position in influence score, The Guardian is a prominent media outlet [2]
    - Sky News is ranked 6th in influence score among UK media outlets [2]

1.3.3 How news outlet and articles are selected.

News outlets for this analysis are carefully selected based on their ratings and representation among their peers. This selection criterion ensures that the chosen outlets accurately reflect the country's perspectives and public discourse. High-rated news outlets are more likely to influence and represent the national sentiment, making their articles valuable for this comparative study.

To achieve a comprehensive and accurate representation of word usage, articles are scraped from various categories including "features," "local news," "opinion pieces," and "sports." This diverse selection captures a wide range of topics and writing styles, enabling a thorough comparison between local and global news outlets. By including multiple categories, we ensure that the analysis covers different types of content, from factual reporting to subjective opinions, thus providing a holistic view of the language and themes prevalent in the news of each country

This method of selection and categorising them for more accurate comparison and insight into the word usage between local and global news coverage

1.4 Ethical considerations

1.4.1 Ethical consideration on articles

Ethical consideration have been considered, making sure each of the website ensuring all links that had been scrapped from each of the website have not been disallowed under the robot.txt of each site.

1.4.2 Potential issue that may result from this analysis.

Due to the nature of a smaller data sample size, data could be skewed from multiple outlier data that have been scraped, causing inaccurate analysis and outcome due to unusually high word count or an atypical usage of certain keywords. Moreover, small sample sizes limit the ability to generalize findings to the broader population. The variability introduced by outliers becomes more pronounced, reducing the statistical power of the analysis and increasing the margin of error. This lack of representativeness and robustness can undermine the reliability and validity of the research outcomes.

Therefore, it is crucial to acknowledge and account for the limitations imposed by small sample sizes in data analysis.

2. Webscraping

2.1 Preparing requirements for web scraping

In [12]:
```
import os
import requests
from bs4 import BeautifulSoup
import pandas as pd
from urllib.parse import urljoin

# Specify headers to accept gzip encoding
headers = {
    'Accept-Encoding': 'gzip, deflate',
    'User-Agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML,
}
```

2.2 Defining scraping and extraction functions

Local

Scraping article links from The Straits Time webpage

In [11]:
```
# Sections to scrape
sections = ['housing', 'health', 'transport', 'politics']

# Base URL template for the website
base_url_template = 'https://www.straitstimes.com/singapore/{}?page='

# CSV file to save the extracted URLs
csv_file = 'scrape_link/local/strait_times_links.csv'

# Function to scrape a single page
def scrape_page(url):
    response = requests.get(url, headers=headers)

    if response.status_code == 200:
        print(f'Successfully fetched {url}')
        soup = BeautifulSoup(response.content, 'html.parser')
```

```
        identifier = soup.find_all(class_='stretched-link')
        urls = [urljoin('https://www.straitstimes.com', link.get('href')) for link in id
        return urls
    else:
        print(f'Failed to fetch {url}. Status code: {response.status_code}')
        return []

# Function to save URLs to CSV
def save_to_csv(urls, file_name):
    urls_df = pd.DataFrame(urls, columns=['URL'])
    if os.path.isfile(file_name):
        urls_df.to_csv(file_name, mode='a', header=False, index=False)
    else:
        urls_df.to_csv(file_name, mode='w', header=True, index=False)

# Loop through sections and pages to scrape URLs
for section in sections:
    base_url = base_url_template.format(section)
    for page_number in range(1, 4):  # Change the range as needed
        page_url = base_url + str(page_number)
        urls = scrape_page(page_url)
        if urls:
            save_to_csv(urls, csv_file)

print('Extracted URLs have been saved to', csv_file)
```

```
Successfully fetched https://www.straitstimes.com/singapore/housing?page=1
Successfully fetched https://www.straitstimes.com/singapore/housing?page=2
Successfully fetched https://www.straitstimes.com/singapore/housing?page=3
Successfully fetched https://www.straitstimes.com/singapore/health?page=1
Successfully fetched https://www.straitstimes.com/singapore/health?page=2
Successfully fetched https://www.straitstimes.com/singapore/health?page=3
Successfully fetched https://www.straitstimes.com/singapore/transport?page=1
Successfully fetched https://www.straitstimes.com/singapore/transport?page=2
Successfully fetched https://www.straitstimes.com/singapore/transport?page=3
Successfully fetched https://www.straitstimes.com/singapore/politics?page=1
Successfully fetched https://www.straitstimes.com/singapore/politics?page=2
Successfully fetched https://www.straitstimes.com/singapore/politics?page=3
Extracted URLs have been saved to scrape_link/local/strait_times_links.csv
```

Scraping article links from Today Online webpage

In [3]:

```
# Sections to scrape
sections = ['features', 'big-read', 'health', 'voices']

# Base URL template for the website
base_url_template = 'https://www.todayonline.com/{}'

# CSV file to save the extracted URLs
csv_file = 'scrape_link/local/today_online_links.csv'

# Function to scrape a single section
def scrape_section(section):
    url = base_url_template.format(section)
    response = requests.get(url, headers=headers)

    if response.status_code == 200:
        print(f'Successfully fetched section {section}')
        soup = BeautifulSoup(response.content, 'html.parser')
        identifier = soup.find_all(class_='h4__link list-object__heading-link')
        urls = [urljoin('https://www.todayonline.com', link.get('href')) for link in ide
        return urls
    else:
        print(f'Failed to fetch {section}. Status code: {response.status_code}')
        return []
```

```
# Function to save URLs to CSV
def save_to_csv(urls, file_name):
    urls_df = pd.DataFrame(urls, columns=['URL'])
    if os.path.isfile(file_name):
        urls_df.to_csv(file_name, mode='a', header=False, index=False)
    else:
        urls_df.to_csv(file_name, mode='w', header=True, index=False)

# Loop through sections to scrape URLs
for section in sections:
    urls = scrape_section(section)
    if urls:
        save_to_csv(urls, csv_file)

print('Extracted URLs have been saved to', csv_file)
```

```
Successfully fetched section features
Successfully fetched section big-read
Successfully fetched section health
Successfully fetched section voices
Extracted URLs have been saved to scrape_link/local/today_online_links.csv
```

Scraping article links from Channel News Asia webpage

In [30]:

```
# Sections to scrape
sections = ['asia', 'singapore', 'sustainability', 'lifestyle', 'sport', 'world']

# Base URL template for the website
base_url_template = 'https://www.channelnewsasia.com/{}'

# CSV file to save the extracted URLs
csv_file = 'scrape_link/local/cna_links.csv'

# Function to scrape a single section
def scrape_section(section):
    url = base_url_template.format(section)
    response = requests.get(url, headers=headers)

    if response.status_code == 200:
        print(f'Successfully fetched section {section}')
        soup = BeautifulSoup(response.content, 'html.parser')
        identifier = soup.find_all(class_='h6__link list-object__heading-link')
        urls = [urljoin('https://www.channelnewsasia.com', link.get('href')) for link in
        return urls
    else:
        print(f'Failed to fetch {section}. Status code: {response.status_code}')
        return []

# Function to save URLs to CSV
def save_to_csv(urls, file_name):
    urls_df = pd.DataFrame(urls, columns=['URL'])
    if os.path.isfile(file_name):
        urls_df.to_csv(file_name, mode='a', header=False, index=False)
    else:
        urls_df.to_csv(file_name, mode='w', header=True, index=False)

# Loop through sections to scrape URLs
for section in sections:
    urls = scrape_section(section)
    if urls:
        save_to_csv(urls, csv_file)

print('Extracted URLs have been saved to', csv_file)
```

```
Successfully fetched section asia
Successfully fetched section singapore
Successfully fetched section sustainability
Successfully fetched section lifestyle
Successfully fetched section sport
Successfully fetched section world
Extracted URLs have been saved to scrape_link/local/cna_links.csv
```

Global

Scraping article links from The Daily Mail webpage

In [23]:

```python
# Sections to scrape
sections = ['news','sport','health']
# sections = ['news']

# Base URL template for the website
base_url_template = 'https://www.dailymail.co.uk/{}/index.html'

# CSV file to save the extracted URLs
csv_file = 'scrape_link/uk/the_dailymail_links.csv'

# Headers to mimic a browser visit
headers = {
    'User-Agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML,
}

# Function to scrape a single section
def scrape_section(section):
    url = base_url_template.format(section)
    response = requests.get(url, headers=headers)

    if response.status_code == 200:
        print(f'Successfully fetched section {section}')
        soup = BeautifulSoup(response.content, 'html.parser')
        # Find all anchor tags with an href attribute
        links = soup.find_all('a', href=True)
        # Use a set to avoid duplicate URLs
        urls = {urljoin('https://www.dailymail.co.uk/', link['href']) for link in links
        return list(urls)[:35]  # Limit to the first 35 URLs
    else:
        print(f'Failed to fetch {section}. Status code: {response.status_code}')
        return []

# Function to save URLs to CSV
def save_to_csv(urls, file_name):
    if not urls:
        return

    urls_df = pd.DataFrame(urls, columns=['URL'])
    if os.path.isfile(file_name):
        urls_df.to_csv(file_name, mode='a', header=False, index=False)
    else:
        urls_df.to_csv(file_name, mode='w', header=True, index=False)

# Ensure the directory exists
os.makedirs(os.path.dirname(csv_file), exist_ok=True)

# Loop through sections to scrape URLs
all_urls = set()
for section in sections:
    urls = scrape_section(section)
    all_urls.update(urls)
```

```
# Save all unique URLs to CSV
save_to_csv(all_urls, csv_file)

print('Extracted URLs have been saved to', csv_file)
```

```
Successfully fetched section news
Successfully fetched section sport
Successfully fetched section health
Extracted URLs have been saved to scrape_link/uk/the_dailymail_links.csv
```

Scraping article links from The Guardian webpage

In [21]:

```
# Sections to scrape
sections = ['international','uk/sport']

# Base URL template for the website
base_url_template = 'https://www.theguardian.com/{}'

# CSV file to save the extracted URLs
csv_file = 'scrape_link/uk/the_guardian_links.csv'

# Function to scrape a single section
def scrape_section(section):
    url = base_url_template.format(section)
    response = requests.get(url, headers=headers)

    if response.status_code == 200:
        print(f'Successfully fetched section {section}')
        soup = BeautifulSoup(response.content, 'html.parser')
        # Find all anchor tags with an href attribute
        links = soup.find_all('a', href=True)
        # Use a set to avoid duplicate URLs
        urls = {urljoin('https://www.theguardian.com/', link['href']) for link in links
        return urls
    else:
        print(f'Failed to fetch {section}. Status code: {response.status_code}')
        return set()

# Function to save URLs to CSV
def save_to_csv(urls, file_name):
    if not urls:
        return

    urls_df = pd.DataFrame(urls, columns=['URL'])
    if os.path.isfile(file_name):
        urls_df.to_csv(file_name, mode='a', header=False, index=False)
    else:
        urls_df.to_csv(file_name, mode='w', header=True, index=False)

# Ensure the directory exists
os.makedirs(os.path.dirname(csv_file), exist_ok=True)

# Loop through sections to scrape URLs
all_urls = set()
for section in sections:
    urls = scrape_section(section)
    all_urls.update(urls)

# Save all unique URLs to CSV
save_to_csv(all_urls, csv_file)

print('Extracted URLs have been saved to', csv_file)
```

```
Successfully fetched section international
Successfully fetched section uk/sport
```

```
Extracted URLs have been saved to scrape_link/uk/the_guardian_links.csv
```

Scraping article links from Sky News webpage

In [22]:

```python
# Sections to scrape
sections = ['uk','world','technology']

# Base URL template for the website
base_url_template = 'https://news.sky.com//{}'

# CSV file to save the extracted URLs
csv_file = 'scrape_link/uk/sky_news_links.csv'

# Function to scrape a single section
def scrape_section(section):
    url = base_url_template.format(section)
    response = requests.get(url, headers=headers)

    if response.status_code == 200:
        print(f'Successfully fetched section {section}')
        soup = BeautifulSoup(response.content, 'html.parser')
        # Find all anchor tags with an href attribute
        links = soup.find_all('a', href=True)
        # Use a set to avoid duplicate URLs
        urls = {urljoin('https://news.sky.com/', link['href']) for link in links if '/st
        return urls
    else:
        print(f'Failed to fetch {section}. Status code: {response.status_code}')
        return set()

# Function to save URLs to CSV
def save_to_csv(urls, file_name):
    if not urls:
        return

    urls_df = pd.DataFrame(urls, columns=['URL'])
    if os.path.isfile(file_name):
        urls_df.to_csv(file_name, mode='a', header=False, index=False)
    else:
        urls_df.to_csv(file_name, mode='w', header=True, index=False)

# Ensure the directory exists
os.makedirs(os.path.dirname(csv_file), exist_ok=True)

# Loop through sections to scrape URLs
all_urls = set()
for section in sections:
    urls = scrape_section(section)
    all_urls.update(urls)

# Save all unique URLs to CSV
save_to_csv(all_urls, csv_file)

print('Extracted URLs have been saved to', csv_file)
```

```
Successfully fetched section uk
Successfully fetched section world
Successfully fetched section technology
Extracted URLs have been saved to scrape_link/uk/sky_news_links.csv
```

2.3 Retrieving article from scraped links.

Local

Scraping article from The Straits Time link

In [14]:
```
# Input CSV file containing URLs
input_csv_file = 'scrape_link/local/strait_times_links.csv'
# Output CSV file to save the scraped articles
output_csv_file = 'scraped_articles/local/scraped_articles_strait_times.csv'

# Ensure the output directory exists
os.makedirs(os.path.dirname(output_csv_file), exist_ok=True)

# Function to scrape the article
def scrape_article(url):
    try:
        response = requests.get(url, headers=headers)
        response.raise_for_status()  # Raise HTTPError for bad responses (4xx or 5xx)

        soup = BeautifulSoup(response.content, 'html.parser')

        # Find the main content of the article
        article_body = soup.find('div', class_='clearfix text-formatted field field--nam

        if article_body:
            paragraphs = article_body.find_all('p')
            article_text = "\n\n".join([para.get_text() for para in paragraphs])
            return article_text
        else:
            print('Failed to find the article content for URL:', url)
            return ''
    except requests.exceptions.RequestException as e:
        print(f'Error fetching the article page {url}: {e}')
        return ''

# Read URLs from the CSV file
df_urls = pd.read_csv(input_csv_file)

# Check if 'URL' column exists
count = 0

if 'URL' not in df_urls.columns:
    print(f'No "URL" column found in {input_csv_file}')
else:
    urls = df_urls['URL'].tolist()

    # List to store all articles
    articles = []

    # Loop through each URL to scrape the article content
    for idx, url in enumerate(urls):
        article_content = scrape_article(url)

        if article_content:
            articles.append({'URL': url, 'Content': article_content})
            print(f'Scraped article from {url}')
            count+=1

    # Save all scraped articles to a single CSV file
    df_articles = pd.DataFrame(articles)
    df_articles.to_csv(output_csv_file, index=False, encoding='utf-8')

    print(f'All articles have been scraped and saved to {output_csv_file}, a total of {c
```

```
Scraped article from https://www.straitstimes.com/singapore/housing/open-concept-hdb-fla
ts-without-partitions-or-beams-to-be-piloted-in-kallang-whampoa-bto-project
Scraped article from https://www.straitstimes.com/singapore/housing/time-is-running-shor
t-chempaka-residents-unsure-about-future-with-leases-expiring-in-10-years
```

Scraped article from https://www.straitstimes.com/singapore/housing/transactions-mostly-profitable-for-chempaka-landed-homes-despite-leases-running-out
Scraped article from https://www.straitstimes.com/singapore/housing/singapore-private-homes-still-most-expensive-in-asia-pacific-hdb-flats-most-attainable-report
Scraped article from https://www.straitstimes.com/singapore/housing/condo-rents-dip-04-in-april-hdb-rents-inch-up-03
Scraped article from https://www.straitstimes.com/singapore/housing/sellers-welcome-ability-to-market-resale-flats-on-hdb-portal-without-help-of-agents
Scraped article from https://www.straitstimes.com/singapore/housing/sellers-can-list-their-flats-on-hdb-flat-portal-from-may-13
Scraped article from https://www.straitstimes.com/singapore/housing/what-you-need-to-know-about-hdb-s-resale-flat-listing-service
Scraped article from https://www.straitstimes.com/singapore/housing/authorities-call-out-2m-hdb-flat-listings-including-misleading-ad-for-jumbo-flat-in-sengkang
Scraped article from https://www.straitstimes.com/singapore/housing/hdb-resale-prices-up-09-in-april-volume-rebounds-157
Scraped article from https://www.straitstimes.com/singapore/housing/new-residential-estate-in-sengkang-could-have-10000-homes
Scraped article from https://www.straitstimes.com/singapore/housing/property-market-to-continue-stabilising-despite-hdb-private-home-prices-rising-in-q1-desmond-lee
Scraped article from https://www.straitstimes.com/singapore/housing/hdb-resale-prices-climb-for-16th-straight-quarter-in-q1-more-flats-sold
Scraped article from https://www.straitstimes.com/singapore/housing/condo-resale-prices-inch-up-in-march-as-volume-rebounds-174
Scraped article from https://www.straitstimes.com/singapore/housing/s-pore-developer-criticised-for-plans-to-turn-170-year-old-hotel-in-adelaide-into-student-housing
Scraped article from https://www.straitstimes.com/singapore/housing/askst-how-will-raising-annual-value-bands-for-owner-occupied-homes-lower-property-tax-bills
Scraped article from https://www.straitstimes.com/singapore/housing/hdb-resale-prices-rise-03-in-march-fewer-units-sold
Scraped article from https://www.straitstimes.com/singapore/housing/hdb-resale-prices-up-17-in-q1-higher-than-11-rise-in-previous-quarter
Scraped article from https://www.straitstimes.com/singapore/housing/s-poreans-who-helped-stranded-malaysians-amid-covid-19-welcome-emergency-housing-portal
Scraped article from https://www.straitstimes.com/singapore/housing/sp-group-offers-extra-6-months-warranty-for-tengah-s-cooling-system-after-leak-complaints
Scraped article from https://www.straitstimes.com/singapore/housing/families-awaiting-bto-flats-say-300-rental-voucher-inadequate-young-couples-welcome-lower-deposit
Scraped article from https://www.straitstimes.com/singapore/housing/new-online-portal-to-help-people-who-need-emergency-housing-during-national-crises
Scraped article from https://www.straitstimes.com/singapore/housing/supermarket-and-food-court-to-open-end-q2-2024-in-first-neighbourhood-centre-in-tengah
Scraped article from https://www.straitstimes.com/singapore/housing/daikin-triples-workers-addressing-cooling-system-defects-after-some-tengah-residents-face-leaks
Scraped article from https://www.straitstimes.com/singapore/housing/switch-from-sibor-to-sora-before-april-30-act-now-more-time-to-choose-home-loan-abs-the-association-of-banks-in-singapore
Scraped article from https://www.straitstimes.com/singapore/housing/four-room-flats-in-hougang-tanglin-halt-most-popular-among-first-timers-in-february-bto-exercise
Scraped article from https://www.straitstimes.com/singapore/housing/condo-resale-prices-dip-08-in-january-after-5-straight-months-of-growth
Scraped article from https://www.straitstimes.com/singapore/housing/5052-million-in-cpf-housing-grants-given-out-to-10400-households-since-feb-2023-hdb
Scraped article from https://www.straitstimes.com/singapore/housing/underground-bus-interchange-in-bidadari-to-be-completed-by-end-2024-after-three-year-delay
Scraped article from https://www.straitstimes.com/singapore/housing/bto-flats-in-tanjong-rhu-holland-village-to-go-on-sale-in-june
Scraped article from https://www.straitstimes.com/singapore/health/pfizer-singaporeans-want-to-embrace-healthier-lifestyles-after-pandemic-wake-up-call
Scraped article from https://www.straitstimes.com/singapore/health/law-firm-to-represent-more-than-150-cordlife-parents-over-their-children-s-cord-blood
Scraped article from https://www.straitstimes.com/singapore/health/new-technique-reduces-risk-of-radiation-side-effects-for-prostate-cancer-patients
Scraped article from https://www.straitstimes.com/singapore/health/hospitals-in-singapore-making-flexible-shifts-more-accessible-for-nurses
Scraped article from https://www.straitstimes.com/singapore/health/redeveloped-alexandra

-hospital-to-house-emergency-department-around-1300-beds
Scraped article from https://www.straitstimes.com/singapore/health/new-multigene-testing-lab-helps-pinpoint-targeted-treatment-for-cancer-patients-in-s-pore-region
Scraped article from https://www.straitstimes.com/singapore/health/what-should-i-look-out-for-when-choosing-a-mosquito-repellent
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All articles have been scraped and saved to scraped_articles/local/scraped_articles_straits_times.csv, a total of 116 articles was scraped.
```

Scraping article from Today Online link

In [15]:

```python
# Input CSV file containing URLs
input_csv_file = 'scrape_link/local/today_online_links.csv'
# Output CSV file to save the scraped articles
output_csv_file = 'scraped_articles/local/scraped_articles_today.csv'

# Ensure the output directory exists
os.makedirs(os.path.dirname(output_csv_file), exist_ok=True)

# Function to scrape the article
def scrape_article(url):
    try:
        response = requests.get(url, headers=headers)
        response.raise_for_status()  # Raise HTTPError for bad responses (4xx or 5xx)

        soup = BeautifulSoup(response.content, 'html.parser')

        # Find the main content of the article
        article_body = soup.find('div', class_='content-wrapper')

        if article_body:
```

```
            paragraphs = article_body.find_all('p')
            article_text = "\n\n".join([para.get_text() for para in paragraphs])
            return article_text
        else:
            print('Failed to find the article content for URL:', url)
            return ''
    except requests.exceptions.RequestException as e:
        print(f'Error fetching the article page {url}: {e}')
        return ''

# Read URLs from the CSV file
df_urls = pd.read_csv(input_csv_file)

# Check if 'URL' column exists
count = 0

if 'URL' not in df_urls.columns:
    print(f'No "URL" column found in {input_csv_file}')
else:
    urls = df_urls['URL'].tolist()

    # List to store all articles
    articles = []

    # Loop through each URL to scrape the article content
    for idx, url in enumerate(urls):
        article_content = scrape_article(url)

        if article_content:
            articles.append({'URL': url, 'Content': article_content})
            print(f'Scraped article from {url}')
            count+=1

    # Save all scraped articles to a single CSV file
    df_articles = pd.DataFrame(articles)
    df_articles.to_csv(output_csv_file, index=False, encoding='utf-8')

    print(f'All articles have been scraped and saved to {output_csv_file}, a total of {c
```

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Scraped article from https://www.todayonline.com/features/love-dad-become-dad-39-work-home-artist-employees-2440636
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Scraped article from https://www.todayonline.com/voices/community-heroes-i-gave-free-music-lessons-underserved-kids-over-time-we-became-close-knit-family-2415846
Scraped article from https://www.todayonline.com/voices/your-say-holding-commonwealth-games-across-several-cities-once-could-be-key-keeping-event-alive-2400246
Scraped article from https://www.todayonline.com/voices/no-degree-required-graveyard-shift-smrt-track-access-rewarding-work-2381126
Scraped article from https://www.todayonline.com/voices/no-degree-required-smrt-train-captain-2355116
Scraped article from https://www.todayonline.com/voices/no-degree-required-how-i-became-hairdresser-leading-team-24-despite-my-familys-initial-concerns-2350286
Scraped article from https://www.todayonline.com/voices/your-say-why-singapore-should-strengthen-laws-protecting-retrenched-workers-2349576
Scraped article from https://www.todayonline.com/voices/no-degree-required-why-im-cool-doing-blue-collar-work-and-servicing-air-cons-2345941
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Scraped article from https://www.todayonline.com/voices/no-degree-required-vinyl-car-wrap-woman-chauvinism-prejudice-manual-work-2333586
Scraped article from https://www.todayonline.com/voices/no-degree-required-i-gave-my-engineering-dreams-now-savour-success-pastry-chef-2331236
Scraped article from https://www.todayonline.com/voices/no-degree-required-landscaping-work-outdoor-love-plants-2326541
Scraped article from https://www.todayonline.com/voices/no-degree-required-electric-vehicles-robots-2321421
Scraped article from https://www.todayonline.com/voices/your-say-funeral-package-pitfalls-2321306
Scraped article from https://www.todayonline.com/voices/no-degree-required-i-have-engineering-diploma-found-my-true-calling-chef-2315591
All articles have been scraped and saved to scraped_articles/local/scraped_articles_today.csv, a total of 75 articles was scraped.
```

Scraping article from Channel News Asia link

In [31]:

```
# Input CSV file containing URLs
input_csv_file = 'scrape_link/local/cna_links.csv'
# Output CSV file to save the scraped articles
```

```
output_csv_file = 'scraped_articles/local/scraped_articles_cna.csv'

# Ensure the output directory exists
os.makedirs(os.path.dirname(output_csv_file), exist_ok=True)

# Function to scrape the article
def scrape_article(url):
    try:
        response = requests.get(url, headers=headers)
        response.raise_for_status()  # Raise HTTPError for bad responses (4xx or 5xx)

        soup = BeautifulSoup(response.content, 'html.parser')

        # Find the main content of the article
        article_body = soup.find('section', class_='block block-layout-builder block-fie

        if article_body:
            paragraphs = article_body.find_all('p')
            article_text = "\n\n".join([para.get_text() for para in paragraphs])
            return article_text
        else:
            print('Failed to find the article content for URL:', url)
            return ''
    except requests.exceptions.RequestException as e:
        return ''

# Read URLs from the CSV file
df_urls = pd.read_csv(input_csv_file)

# Check if 'URL' column exists
count = 0

if 'URL' not in df_urls.columns:
    print(f'No "URL" column found in {input_csv_file}')
else:
    urls = df_urls['URL'].tolist()

    # List to store all articles
    articles = []

    # Loop through each URL to scrape the article content
    for idx, url in enumerate(urls):
        article_content = scrape_article(url)

        if article_content:
            articles.append({'URL': url, 'Content': article_content})
            print(f'Scraped article from {url}')
            count+=1

    # Save all scraped articles to a single CSV file
    df_articles = pd.DataFrame(articles)
    df_articles.to_csv(output_csv_file, index=False, encoding='utf-8')

    print(f'All articles have been scraped and saved to {output_csv_file}, a total of {c
```

```
Scraped article from https://www.channelnewsasia.com/asia/philippine-vp-sara-duterte-qui
ts-marcos-cabinet-alliance-4421926
Scraped article from https://www.channelnewsasia.com/asia/malaysia-ulu-tiram-police-stat
ion-attack-family-members-terrorism-islamic-state-4420996
Scraped article from https://www.channelnewsasia.com/asia/russia-vietnam-vladimir-putin-
influence-southeast-asia-defence-4421081
Scraped article from https://www.channelnewsasia.com/asia/vietnam-leadership-reshuffle-c
orruption-dinh-tien-dung-step-down-hanoi-4421626
Scraped article from https://www.channelnewsasia.com/asia/indonesia-online-gambling-outf
low-funds-southeast-asia-illegal-money-4420811
Scraped article from https://www.channelnewsasia.com/east-asia/china-tourists-uncivilise
```

d-behaviour-chengdu-panda-research-base-ban-life-4421276
Scraped article from https://www.channelnewsasia.com/sport/new-asian-champions-earn-us12-million-prize-money-club-competition-revamp-4421071
Scraped article from https://www.channelnewsasia.com/asia/malaysia-china-50-years-ties-trade-deal-durian-economic-development-4421131
Scraped article from https://www.channelnewsasia.com/east-asia/taiwan-president-lai-china-pressure-will-not-bow-4420801
Scraped article from https://www.channelnewsasia.com/sport/eleven-swimmers-doping-scandal-named-china-olympic-team-4420806
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Scraped article from https://www.channelnewsasia.com/asia/court-finds-four-philippine-police-guilty-drug-war-killings-manila-rodrigo-duterte-4418246
Scraped article from https://www.channelnewsasia.com/asia/malaysia-brics-emerging-economies-trade-alliance-anwar-ibrahim-4418001
Scraped article from https://www.channelnewsasia.com/commentary/putin-russia-north-korea-ties-kim-jong-un-balloons-4418301
Scraped article from https://www.channelnewsasia.com/asia/malaysia-anwar-ibrahim-diesel-subsidies-tour-buses-agencies-leakages-4418191
Scraped article from https://www.channelnewsasia.com/asia/malaysia-vep-singapore-drivers-frustration-johor-4407671
Scraped article from https://www.channelnewsasia.com/east-asia/china-japan-south-korea-free-trade-agreement-rcep-plus-4391881
Scraped article from https://www.channelnewsasia.com/asia/x-twitter-porn-policy-violate-law-indonesia-malaysia-4391461
Scraped article from https://www.channelnewsasia.com/asia/indonesia-house-lizards-geckos-export-china-wildlife-trade-traditional-medicine-4388586
Scraped article from https://www.channelnewsasia.com/singapore/hdb-bto-june-2024-exercise-tanjong-rhu-holland-vista-4420781
Scraped article from https://www.channelnewsasia.com/singapore/diver-dies-entangled-boat-propeller-marina-south-pier-mom-4421851
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Scraped article from https://www.channelnewsasia.com/singapore/pub-reservoir-water-bidadari-housing-underground-land-scarce-grace-fu-4419016
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Scraped article from https://www.channelnewsasia.com/singapore/woman-deity-cheated-followers-7-million-eat-faeces-pull-teeth-woo-may-hoe-4420716
Scraped article from https://www.channelnewsasia.com/singapore/man-sexually-violated-mistress-daughter-attempted-suicide-jailed-4421451
Scraped article from https://www.channelnewsasia.com/singapore/oil-spill-east-coast-sentosa-impact-sea-sports-business-long-weekend-revenue-losses-4418216
Scraped article from https://www.channelnewsasia.com/singapore/maid-moonlight-stole-23k-part-time-cleaning-jail-4418496
Scraped article from https://www.channelnewsasia.com/singapore/man-jailed-host-drug-party-guests-covid-4418326
Scraped article from https://www.channelnewsasia.com/wellness/sepsis-bacterial-infection-limb-amputation-4409721
Scraped article from https://www.channelnewsasia.com/business/singtel-kkr-consortium-invest-us13-billion-st-telemedia-global-data-centres-4418801
Scraped article from https://www.channelnewsasia.com/singapore/pisa-singapore-study-creative-thinking-students-4414516
Scraped article from https://www.channelnewsasia.com/singapore/tidal-currents-pose-challenge-oil-spill-clean-operations-drones-satellite-imagery-deployed-help-effort-4418166
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od-delivery-riders-paid-paynow-edited-screenshots-4418146
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h-clash-austria-4421921
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ash-against-albania-4421906
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1-triple-header-4421911
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aunder-us-drug-money-4419996
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eafarer-shortage-4421136
Scraped article from https://www.channelnewsasia.com/world/britains-stonehenge-targed-pa
int-environmental-protesters-4421676
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-serious-concerns-under-laws-war-un-4422436
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limousine-ride-4421651
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hosa-starts-new-term-multi-party-government-4421501
Scraped article from https://www.channelnewsasia.com/sport/track-less-tuvalu-sends-one-m
an-team-paris-olympics-4421046
Scraped article from https://www.channelnewsasia.com/sport/baseball-legend-willie-mays-d
ead-93-4420656
Scraped article from https://www.channelnewsasia.com/world/haj-pilgrims-550-died-heat-te
mperatures-egypt-4419891
All articles have been scraped and saved to scraped_articles/local/scraped_articles_cna.
csv, a total of 92 articles was scraped.
```

Global

Scraping article from The Daily Mail link

In [24]:

```python
# Input CSV file containing URLs
input_csv_file = 'scrape_link/uk/the_dailymail_links.csv'
# Output CSV file to save the scraped articles
output_csv_file = 'scraped_articles/uk/scraped_articles_dailymail.csv'

# Ensure the output directory exists
os.makedirs(os.path.dirname(output_csv_file), exist_ok=True)

# Function to scrape the article
def scrape_article(url):
    try:
        response = requests.get(url, headers=headers)
        response.raise_for_status()  # Raise HTTPError for bad responses (4xx or 5xx)

        soup = BeautifulSoup(response.content, 'html.parser')
```

```
        # Find the main content of the article
        article_body = soup.find('div', itemprop='articleBody')

        if article_body:
            paragraphs = article_body.find_all('p')
            article_text = "\n\n".join([para.get_text() for para in paragraphs])
            return article_text
        else:
            print('Failed to find the article content for URL:', url)
            return ''
    except requests.exceptions.RequestException as e:
        print(f'Error fetching the article page {url}: {e}')
        return ''

# Read URLs from the CSV file
df_urls = pd.read_csv(input_csv_file)

# Check if 'URL' column exists
count = 0

if 'URL' not in df_urls.columns:
    print(f'No "URL" column found in {input_csv_file}')
else:
    urls = df_urls['URL'].tolist()

    # List to store all articles
    articles = []

    # Loop through each URL to scrape the article content
    for idx, url in enumerate(urls):
        article_content = scrape_article(url)

        if article_content:
            articles.append({'URL': url, 'Content': article_content})
            print(f'Scraped article from {url}')
            count+=1

    # Save all scraped articles to a single CSV file
    df_articles = pd.DataFrame(articles)
    df_articles.to_csv(output_csv_file, index=False, encoding='utf-8')

    print(f'All articles have been scraped and saved to {output_csv_file}, a total of {c
```

```
Scraped article from https://www.dailymail.co.uk/news/article-13544315/Damien-Hirst-actr
essSophie-Cannell-pictured-two-prams-one-newborn-dog.html#comments
Scraped article from https://www.dailymail.co.uk/femail/article-13542575/royal-ascot-car
riage-procession-lady-gabriella-thomas-kingston.html#video
Scraped article from https://www.dailymail.co.uk/news/article-13543529/Gary-Lineker-cart
e-blanche-BBC-bosses-turning-blind.html
Scraped article from https://www.dailymail.co.uk/tvshowbiz/article-13542261/Nick-Knowles
-Katie-Dadzie-Royal-Ascot.html
Scraped article from https://www.dailymail.co.uk/health/article-13541641/deaths-rare-bac
terial-infection-Japan-mystify-experts-UK.html#video
Scraped article from https://www.dailymail.co.uk/sport/football/article-13543009/Germany
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injury-debate-helmet-cycling-safer-dangerous.html#video
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tor-cancer-patient-cannabis-treatment.html
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kicks-Paris-Fashion-Week-floral-Chanel-outfit.html
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scover-worlds-oldest-wine.html#comments
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n-Holmes-prostate-cancer.html#comments
Scraped article from https://www.dailymail.co.uk/news/article-13542739/taylor-swift-cardiff-palestine-protesters.html#comments
Scraped article from https://www.dailymail.co.uk/femail/article-13539245/Inside-Prince-Williams-bond-Duchess-Edinburgh-Sophie.html#video
Scraped article from https://www.dailymail.co.uk/shopping-us/article-13501613/rodial-vitamin-C-pads-brighter-skin-discount-code.html
Scraped article from https://www.dailymail.co.uk/sport/football/article-13535873/Holland-fans-accused-doing-blackface-attempt-dress-legend-Ruud-Gullit-Euro-2024-victory-Poland.html
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g-Whitley-Jason-Statham-children-helicopter.html
All articles have been scraped and saved to scraped_articles/uk/scraped_articles_dailyma
il.csv, a total of 104 articles was scraped.
```

Scraping article from The Guardian link

In [25]:

```python
# Input CSV file containing URLs
input_csv_file = 'scrape_link/uk/the_guardian_links.csv'
# Output CSV file to save the scraped articles
output_csv_file = 'scraped_articles/uk/scraped_articles_guardian.csv'

# Ensure the output directory exists
os.makedirs(os.path.dirname(output_csv_file), exist_ok=True)

# Function to scrape the article
def scrape_article(url):
    try:
        response = requests.get(url, headers=headers)
        response.raise_for_status()  # Raise HTTPError for bad responses (4xx or 5xx)

        soup = BeautifulSoup(response.content, 'html.parser')

        # Find the main content of the article
        article_body = soup.find('div', class_='article-body-commercial-selector')

        if article_body:
            paragraphs = article_body.find_all('p')
            article_text = "\n\n".join([para.get_text() for para in paragraphs])
            return article_text
        else:
            print('Failed to find the article content for URL:', url)
            return ''
    except requests.exceptions.RequestException as e:
        print(f'Error fetching the article page {url}: {e}')
        return ''

# Read URLs from the CSV file
df_urls = pd.read_csv(input_csv_file)

# Check if 'URL' column exists
count = 0

if 'URL' not in df_urls.columns:
    print(f'No "URL" column found in {input_csv_file}')
else:
    urls = df_urls['URL'].tolist()

    # List to store all articles
    articles = []
```

```
    # Loop through each URL to scrape the article content
    for idx, url in enumerate(urls):
        article_content = scrape_article(url)

        if article_content:
            articles.append({'URL': url, 'Content': article_content})
            print(f'Scraped article from {url}')
            count+=1

    # Save all scraped articles to a single CSV file
    df_articles = pd.DataFrame(articles)
    df_articles.to_csv(output_csv_file, index=False, encoding='utf-8')

    print(f'All articles have been scraped and saved to {output_csv_file}, a total of {c
```

```
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Scraped article from https://www.theguardian.com/world/article/2024/jun/18/hundreds-of-hajj-pilgrims-die-in-mecca-from-heat-related-illness
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All articles have been scraped and saved to scraped_articles/scraped_articles_guardian.csv, a total of 139 articles was scraped.
```

Scraping article from Sky News link

In [26]:

```python
# Input CSV file containing URLs
input_csv_file = 'scrape_link/uk/sky_news_links.csv'
# Output CSV file to save the scraped articles
output_csv_file = 'scraped_articles/uk/scraped_articles_skynews.csv'

# Ensure the output directory exists
os.makedirs(os.path.dirname(output_csv_file), exist_ok=True)

# Function to scrape the article
def scrape_article(url):
    try:
        response = requests.get(url, headers=headers)
        response.raise_for_status()  # Raise HTTPError for bad responses (4xx or 5xx)

        soup = BeautifulSoup(response.content, 'html.parser')

        # Find the main content of the article
        article_body = soup.find('div', class_='sdc-article-body sdc-article-body--story

        if article_body:
            paragraphs = article_body.find_all('p')
            article_text = "\n\n".join([para.get_text() for para in paragraphs])
            return article_text
        else:
            print('Failed to find the article content for URL:', url)
            return ''
    except requests.exceptions.RequestException as e:
        print(f'Error fetching the article page {url}: {e}')
        return ''
```

```python
# Read URLs from the CSV file
df_urls = pd.read_csv(input_csv_file)

# Check if 'URL' column exists
count = 0

if 'URL' not in df_urls.columns:
    print(f'No "URL" column found in {input_csv_file}')
else:
    urls = df_urls['URL'].tolist()

    # List to store all articles
    articles = []

    # Loop through each URL to scrape the article content
    for idx, url in enumerate(urls):
        article_content = scrape_article(url)

        if article_content:
            articles.append({'URL': url, 'Content': article_content})
            print(f'Scraped article from {url}')
            count+=1

    # Save all scraped articles to a single CSV file
    df_articles = pd.DataFrame(articles)
    df_articles.to_csv(output_csv_file, index=False, encoding='utf-8')

    print(f'All articles have been scraped and saved to {output_csv_file}, a total of {c
```

```
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Scraped article from https://news.sky.com/story/a-lucky-bet-and-unlimited-coffee-how-nvidia-become-the-worlds-most-valuable-company-13155344
Scraped article from https://news.sky.com/story/controversial-microsoft-ai-screenshot-feature-delayed-over-security-concerns-13152918
Scraped article from https://news.sky.com/story/tariffs-hurt-consumers-and-invite-retaliation-uk-should-be-wary-of-following-eus-plans-for-china-13152606

Scraped article from https://news.sky.com/story/dont-know-who-to-vote-for-heres-a-very-simple-guide-to-what-each-party-is-promising-13155274
Scraped article from https://news.sky.com/story/children-in-the-uk-are-shorter-fatter-and-sicker-due-to-poor-diet-and-poverty-report-suggests-13155257
Scraped article from https://news.sky.com/story/tesla-cybertruck-buyers-reporting-delivery-delays-over-windscreen-wipers-issue-13154258
Scraped article from https://news.sky.com/story/love-island-star-jack-fowler-calls-for-nut-ban-on-planes-after-extremely-scary-allergic-reaction-13155381
Scraped article from https://news.sky.com/story/nasa-accidentally-broadcasts-simulation-of-astronaut-in-distress-13152304
Scraped article from https://news.sky.com/story/vermont-republican-representative-sorry-after-being-filmed-pouring-water-in-democratic-colleagues-bag-13155179
Scraped article from https://news.sky.com/story/apple-to-integrate-ai-including-chatgpt-into-new-phones-as-elon-musk-threatens-ban-over-security-fears-13151034
Scraped article from https://news.sky.com/story/taylor-swift-impresses-in-cardiff-as-she-greets-fans-in-welsh-13155358
Scraped article from https://news.sky.com/story/will-artificial-intelligence-make-people-buy-apple-products-again-13151397
Scraped article from https://news.sky.com/story/nvidia-overtakes-microsoft-to-become-worlds-most-valuable-public-company-13155098
Scraped article from https://news.sky.com/story/euro-2024-france-captain-kylian-mbappe-suffers-broken-nose-during-victory-opener-against-austria-13154667
Scraped article from https://news.sky.com/story/kay-burley-emily-maitlis-and-mishal-husain-feature-on-tatler-front-cover-as-they-prepare-to-lead-election-coverage-13155307
Scraped article from https://news.sky.com/story/duke-and-duchess-of-edinburgh-share-25th-wedding-anniversary-portrait-13155224
Scraped article from https://news.sky.com/story/manifesto-checker-what-are-the-conservatives-and-liberal-democrats-key-pledges-13150669
Scraped article from https://news.sky.com/story/tories-heading-for-worst-ever-defeat-while-labour-on-track-for-256-majority-poll-finds-13155056
Scraped article from https://news.sky.com/story/labour-candidate-andy-brown-suspended-for-sharing-pro-russian-posts-13155289
Scraped article from https://news.sky.com/story/jordan-bardella-who-is-28-year-old-far-right-leader-who-could-be-frances-next-prime-minister-13150627
Scraped article from https://news.sky.com/story/joey-barton-must-pay-extra-35-000-to-settle-libel-claim-with-jeremy-vine-13155389
Scraped article from https://news.sky.com/story/the-two-theories-behind-putins-north-korea-visit-and-why-south-korea-will-be-particularly-concerned-13155455
Scraped article from https://news.sky.com/story/one-of-the-great-wonders-of-nature-insect-migration-superhighway-discovered-13151698
Scraped article from https://news.sky.com/story/elephants-have-nicknames-for-each-other-scientists-find-in-really-exciting-discovery-13151294
Scraped article from https://news.sky.com/story/jay-slater-former-coronation-street-star-prays-for-safe-return-of-teenager-missing-after-tenerife-festival-13155324
Scraped article from https://news.sky.com/story/scientists-hopeful-of-game-changing-parkinsons-test-able-to-predict-condition-seven-years-before-first-symptoms-13154931
Scraped article from https://news.sky.com/story/putins-visit-to-north-korea-is-a-very-much-a-diplomatic-two-fingers-to-the-west-13154637
Scraped article from https://news.sky.com/story/smart-rings-england-stars-spotted-wearing-device-popular-with-celebrities-including-kim-kardashian-but-can-it-help-them-at-euro-2024-13153093
Scraped article from https://news.sky.com/story/tk-maxx-and-homesense-recall-selected-kintra-teas-that-may-contain-insects-13155338
Scraped article from https://news.sky.com/story/putin-knew-ukraine-would-reject-his-ceasefire-offer-where-will-the-escalation-end-13153051
Scraped article from https://news.sky.com/story/hyper-realistic-humanoid-robots-could-be-used-in-psychotherapy-13151120
Scraped article from https://news.sky.com/story/putins-trip-to-north-korea-underscores-relationship-but-china-will-be-watching-13155363
Scraped article from https://news.sky.com/story/e-coli-outbreak-public-warned-not-to-eat-specific-product-recalled-as-precautionary-measure-13154193
Scraped article from https://news.sky.com/story/at-least-one-dead-in-new-mexico-village-from-wildfire-that-forced-thousands-flee-their-homes-13155219
Scraped article from https://news.sky.com/story/holding-olympics-in-july-will-be-nearly-impossible-over-extreme-heat-report-from-athletes-and-scientists-warns-13154649
Scraped article from https://news.sky.com/story/ai-can-now-detect-if-cats-are-in-pain-by

```
-scanning-feline-faces-13152363
Scraped article from https://news.sky.com/story/migrants-try-to-board-small-boat-off-french-beach-in-desperate-bid-for-britain-as-powerless-police-watch-on-13154980
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Scraped article from https://news.sky.com/story/mcdonalds-ends-ai-drive-thru-trial-after-order-mishaps-13155091
Scraped article from https://news.sky.com/story/tesla-shareholders-approve-elon-musks-huge-pay-package-13152792
Scraped article from https://news.sky.com/story/kim-jong-un-hails-fiery-friendship-with-vladimir-putin-as-russian-president-visits-north-korea-13155292
Scraped article from https://news.sky.com/story/dengue-fever-cases-rising-as-climate-change-draws-tiger-mosquitoes-to-europe-13151795
Failed to find the article content for URL: https://news.sky.com/story/inflation-latest-interest-rates-bank-of-england-money-sky-news-blog-13040934
Scraped article from https://news.sky.com/story/no-one-came-to-help-us-only-eleven-migrants-survived-boat-capsizing-in-one-of-ionian-seas-worst-disasters-13155229
Scraped article from https://news.sky.com/story/e-coli-outbreak-what-are-the-warning-signs-of-infection-13149471
Scraped article from https://news.sky.com/story/government-dropping-windrush-review-recommendations-unlawful-high-court-rules-13155432
All articles have been scraped and saved to scraped_articles/scraped_articles_skynews.csv, a total of 86 articles was scraped.
```

3. Data cleaning and processing

3.1 Preparing requirements for data cleaning

In [5]:
```python
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer
import re

# Download necessary NLTK resources
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
```

```
[nltk_data] Downloading package punkt to
[nltk_data]     C:\Users\Admin\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\Admin\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data]     C:\Users\Admin\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
```

Out[5]:
```
True
```

Cleaning of text data for local news articles

In [8]:
```python
# Load stop words
stop_words = set(stopwords.words('english'))

# Lemmatizer
lemmatizer = WordNetLemmatizer()

def clean_text(text):
    # Lowercase
    text = text.lower()
    # Remove non-alphanumeric characters
```

```
    text = re.sub(r'[^a-zA-Z0-9\s]', '', text)
    # Tokenize text
    tokens = word_tokenize(text)
    # Remove stopwords and lemmatize
    tokens = [lemmatizer.lemmatize(word) for word in tokens if word not in stop_words]
    return ' '.join(tokens)

# List of input file paths
input_files = [
    'scraped_articles/local/scraped_articles_cna.csv',
    'scraped_articles/local/scraped_articles_strait_times.csv',
    'scraped_articles/local/scraped_articles_today.csv'
    # Add more file paths as needed
]

# Output file path
output_csv_file = 'cleaned_data/local/cleaned_articles_combined_local.csv'

# Ensure the output directory exists
os.makedirs(os.path.dirname(output_csv_file), exist_ok=True)

# List to hold cleaned DataFrames
cleaned_data_frames = []

# Process each input file
for input_csv_file in input_files:
    # Read the CSV file
    df = pd.read_csv(input_csv_file)

    # Apply the cleaning function to the 'Content' column
    df['cleaned_article'] = df['Content'].apply(clean_text)

    # Select only the 'URL' and 'cleaned_article' columns
    cleaned_df = df[['cleaned_article']]

    # Append the cleaned DataFrame to the list
    cleaned_data_frames.append(cleaned_df)

    print(f'Processed and cleaned data from {input_csv_file}')

# Concatenate all cleaned DataFrames into a single DataFrame
combined_cleaned_df = pd.concat(cleaned_data_frames, ignore_index=True)

# Save the combined cleaned data to a single CSV file
combined_cleaned_df.to_csv(output_csv_file, index=False)

print(f'Combined cleaned articles have been saved to {output_csv_file}')
```

```
Processed and cleaned data from scraped_articles/local/scraped_articles_cna.csv
Processed and cleaned data from scraped_articles/local/scraped_articles_strait_times.csv
Processed and cleaned data from scraped_articles/local/scraped_articles_today.csv
Combined cleaned articles have been saved to cleaned_data/local/cleaned_articles_combine
d_local.csv
```

Cleaning of text data for global news articles

In [5]:

```
# Load stop words
stop_words = set(stopwords.words('english'))

# Lemmatizer
lemmatizer = WordNetLemmatizer()

def clean_text(text):
    # Lowercase
    text = text.lower()
```

```
    # Remove non-alphanumeric characters
    text = re.sub(r'[^a-zA-Z0-9\s]', '', text)
    # Tokenize text
    tokens = word_tokenize(text)
    # Remove stopwords and lemmatize
    tokens = [lemmatizer.lemmatize(word) for word in tokens if word not in stop_words]
    return ' '.join(tokens)

# List of input file paths
input_files = [
    'scraped_articles/uk/scraped_articles_dailymail.csv',
    'scraped_articles/uk/scraped_articles_guardian.csv',
    'scraped_articles/uk/scraped_articles_skynews.csv'
    # Add more file paths as needed
]

# Output file path
output_csv_file = 'cleaned_data/uk/cleaned_articles_combined_uk.csv'

# Ensure the output directory exists
os.makedirs(os.path.dirname(output_csv_file), exist_ok=True)

# List to hold cleaned DataFrames
cleaned_data_frames = []

# Process each input file
for input_csv_file in input_files:
    # Read the CSV file
    df = pd.read_csv(input_csv_file)

    # Apply the cleaning function to the 'Content' column
    df['cleaned_article'] = df['Content'].apply(clean_text)

    # Select only the 'URL' and 'cleaned_article' columns
    cleaned_df = df[['cleaned_article']]

    # Append the cleaned DataFrame to the list
    cleaned_data_frames.append(cleaned_df)

    print(f'Processed and cleaned data from {input_csv_file}')

# Concatenate all cleaned DataFrames into a single DataFrame
combined_cleaned_df = pd.concat(cleaned_data_frames, ignore_index=True)

# Save the combined cleaned data to a single CSV file
combined_cleaned_df.to_csv(output_csv_file, index=False)

print(f'Combined cleaned articles have been saved to {output_csv_file}')
```

```
Processed and cleaned data from scraped_articles/uk/scraped_articles_dailymail.csv
Processed and cleaned data from scraped_articles/uk/scraped_articles_guardian.csv
Processed and cleaned data from scraped_articles/uk/scraped_articles_skynews.csv
Combined cleaned articles have been saved to cleaned_data/uk/cleaned_articles_combined_u
k.csv
```

3.2 Getting the count of each word from local article

In [35]:

```
import pandas as pd
from collections import Counter

# Path to the cleaned CSV file
input_csv_file = 'cleaned_data/local/cleaned_articles_combined_local.csv'

# Read the CSV file
df = pd.read_csv(input_csv_file)
```

```
# Check if the 'cleaned_article' column exists
if 'cleaned_article' not in df.columns:
    print("The specified column 'cleaned_article' does not exist in the CSV file.")
    exit()

# Concatenate all cleaned articles into a single string
all_text = ' '.join(df['cleaned_article'].dropna())

# Split the text into words
words = all_text.split()

# Count the frequency of each word
word_count = Counter(words)

# Convert the word count dictionary to a DataFrame
word_count_df = pd.DataFrame(list(word_count.items()), columns=['Word', 'Count'])

# Sort the DataFrame by 'Count' in descending order
word_count_df.sort_values(by='Count', ascending=False, inplace=True)

# Path to save the word count CSV
output_word_count_csv = 'cleaned_data/local/word_count_local.csv'

# Save the word count to a CSV file
word_count_df.to_csv(output_word_count_csv, index=False)

# Print the entire sorted list of word frequencies as a table without index
top_20_words = word_count_df.head(20)
print("Full List of Word Frequencies:")
print(top_20_words.to_string(index=False))
print(" ")
print(f"Word count has been saved to {output_word_count_csv}")
```

```
Full List of Word Frequencies:
     Word  Count
     said   1219
singapore    683
     also    517
     year    426
       mr    385
      may    375
     time    348
      one    346
      per    308
     cent    278
    would    250
   people    236
    first    227
    added    225
     work    220
      two    215
    child    212
     home    207
     like    197
      new    196

Word count has been saved to cleaned_data/local/word_count_local.csv
```

Getting the count of each word from UK article

In [34]:

```
import pandas as pd
from collections import Counter

# Path to the cleaned CSV file
```

```
input_csv_file = 'cleaned_data/uk/cleaned_articles_combined_uk.csv'

# Read the CSV file
df = pd.read_csv(input_csv_file)

# Check if the 'cleaned_article' column exists
if 'cleaned_article' not in df.columns:
    print("The specified column 'cleaned_article' does not exist in the CSV file.")
    exit()

# Concatenate all cleaned articles into a single string
all_text = ' '.join(df['cleaned_article'].dropna())

# Split the text into words
words = all_text.split()

# Count the frequency of each word
word_count = Counter(words)

# Convert the word count dictionary to a DataFrame
word_count_df = pd.DataFrame(list(word_count.items()), columns=['Word', 'Count'])

# Sort the DataFrame by 'Count' in descending order
word_count_df.sort_values(by='Count', ascending=False, inplace=True)

# Path to save the word count CSV
output_word_count_csv = 'cleaned_data/uk/word_count_uk.csv'

# Save the word count to a CSV file
word_count_df.to_csv(output_word_count_csv, index=False)

# Print the entire sorted list of word frequencies as a table without index
top_20_words = word_count_df.head(20)
print("Full List of Word Frequencies:")
print(top_20_words.to_string(index=False))
print(" ")
print(f"Word count has been saved to {output_word_count_csv}")
```

```
Full List of Word Frequencies:
   Word  Count
   said   1022
    one    769
   year    683
 people    632
    say    597
  would    569
   also    553
   time    524
  first    516
   like    494
   last    438
    get    432
  world    430
    two    413
   news    411
    day    405
    new    402
  could    393
england    356
   team    347

Word count has been saved to cleaned_data/uk/word_count_uk.csv
```

4. Exploratory data analysis

4.1 Wordclouds

Preparing requirements for data analysis

In [ ]:
```
import pandas as pd
from wordcloud import WordCloud
import matplotlib.pyplot as plt
```

Local

In [23]:
```
# Path to the CSV file with word counts
csv_file_path = 'cleaned_data/local/word_count_local.csv'

# Load data from CSV
df = pd.read_csv(csv_file_path)

# Convert DataFrame to dictionary with words as keys and counts as values
word_freq = dict(zip(df.Word, df.Count))

# Generate a word cloud image
wordcloud = WordCloud(width=800, height=400, background_color='white').generate_from_fre

# Display the generated image using matplotlib
plt.figure(figsize=(10, 5))  # Adjust to appropriate dimensions
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')  # No axis for a cleaner presentation
plt.show()
```


singapore
said
also
one
year
mr
time
would
people
may
two
cent
first
added
cancer
home
child
new
work
like
minister
country
government
national
international
system
water
health
group
centre
service
help
care
school
company
prime
public
around
area
2023
life
flat
need
oil
family
told
take
well
part
three
make
dr
many
get
place
drug
come
could
per
month
day
support
issue
car
m
since
including
made
statement
authority
business
experience
south
stage
last
medical
million
road
state
started
agency
russia
sea
hdb
reported
effort
least
condition
election
price
food
young
friend
according
know
back
much
bus
today
four
think
use
feel
early
study
event
housing
noted
social
start
vehicle
officer
community
body
give
law
six
able
unit
jun
high
week
thing
due
april
used
hour
next
university
park
later
side
took
property
another
without
game
lee
dont
treatment
medium
party
want
risk
process
project
look
data
continue
case
china
parent
general
way
go
period
set
among
member
owner
end
team
wong
hospital
patient
even
u
still
change
resale
ministry
phone
however
already
important
10
2022
2024
malaysia
singaporean
response
development
technology
found
expert
first
cancer
three
make
game
another
without
dr
many
part
well
take
added
told
family
oil
need
two
country
minister
like
would
people
work
mr
time
said
may
also
one
year
cent
could
per
home
water
health
group
centre
service
help
care
school
company
prime
public
around
area
2023
life
flat
need
oil
family
told

```
# Generate a word cloud image
wordcloud = WordCloud(width=800, height=400, background_color='white').generate_from_fre

# Display the generated image using matplotlib
plt.figure(figsize=(10, 5))  # Adjust to appropriate dimensions
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')  # No axis for a cleaner presentation
plt.show()
```



4.2 Comparative Frequency Analysis

In [26]:
```
# Load the data from the provided CSV files
file_path_local = 'cleaned_data/uk/word_count_uk.csv'
file_path_uk = 'cleaned_data/local/word_count_local.csv'

df_local = pd.read_csv(file_path_local)
df_uk = pd.read_csv(file_path_uk)

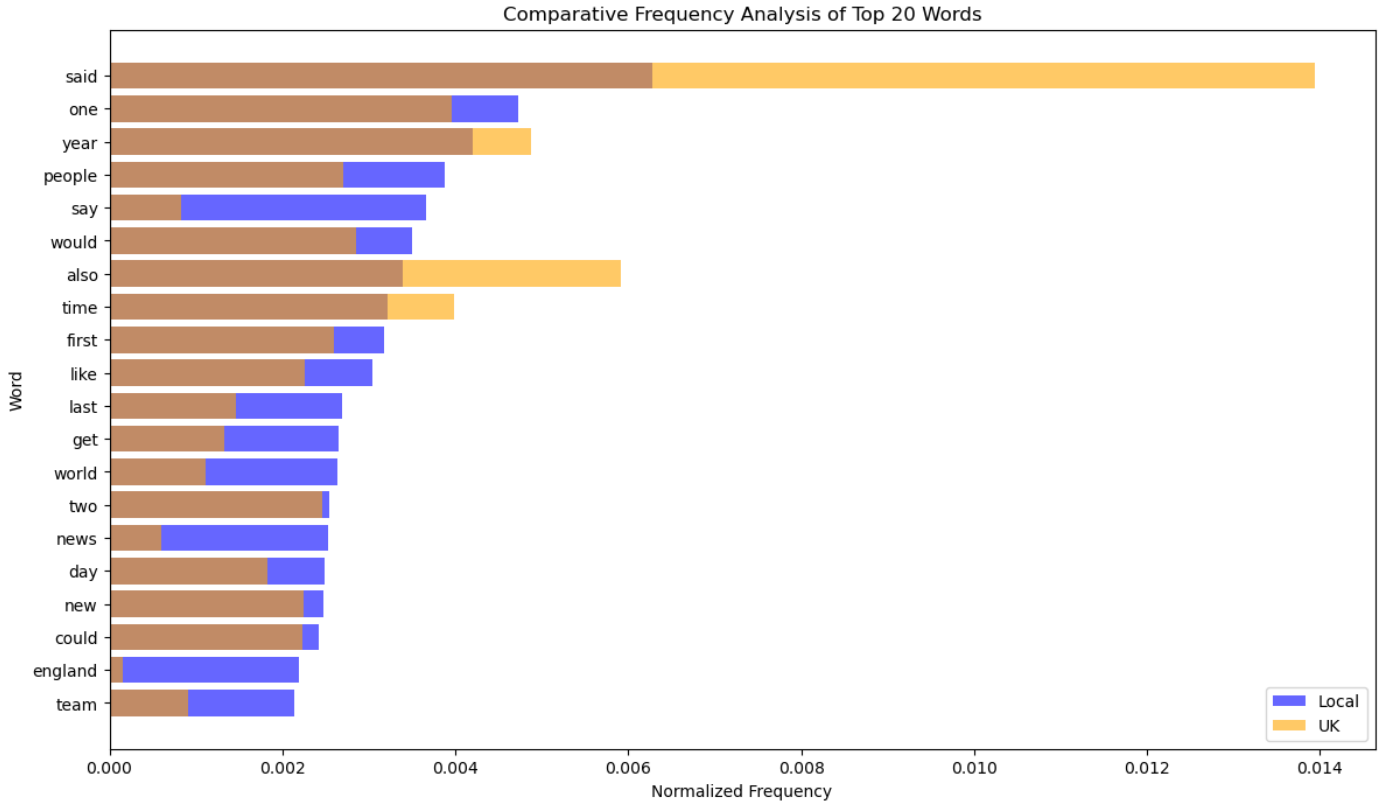
# Merge the dataframes on the 'Word' column
df_merged = pd.merge(df_local, df_uk, on='Word', how='outer', suffixes=('_local', '_uk')

# Calculate the total counts for normalization
total_local = df_local['Count'].sum()
total_uk = df_uk['Count'].sum()

# Normalize the counts
df_merged['Count_local_normalized'] = df_merged['Count_local'] / total_local
df_merged['Count_uk_normalized'] = df_merged['Count_uk'] / total_uk

# Sort by the local normalized count for better visualization
df_merged_sorted = df_merged.sort_values(by='Count_local_normalized', ascending=False).h

# Plot the comparative frequency analysis
plt.figure(figsize=(14, 8))
plt.barh(df_merged_sorted['Word'], df_merged_sorted['Count_local_normalized'], color='bl
plt.barh(df_merged_sorted['Word'], df_merged_sorted['Count_uk_normalized'], color='orang
plt.xlabel('Normalized Frequency')
plt.ylabel('Word')
plt.title('Comparative Frequency Analysis of Top 20 Words')
plt.legend()
plt.gca().invert_yaxis()
plt.show()
```



4.3 Differential Word Analysis

In [31]:

```
# Load the data from the provided CSV files
file_path_local = 'cleaned_data/local/word_count_local.csv'
file_path_uk = 'cleaned_data/uk/word_count_uk.csv'

df_local = pd.read_csv(file_path_local)
df_uk = pd.read_csv(file_path_uk)

# Merge the dataframes on the 'Word' column
df_merged = pd.merge(df_local, df_uk, on='Word', how='outer', suffixes=('_local', '_uk')

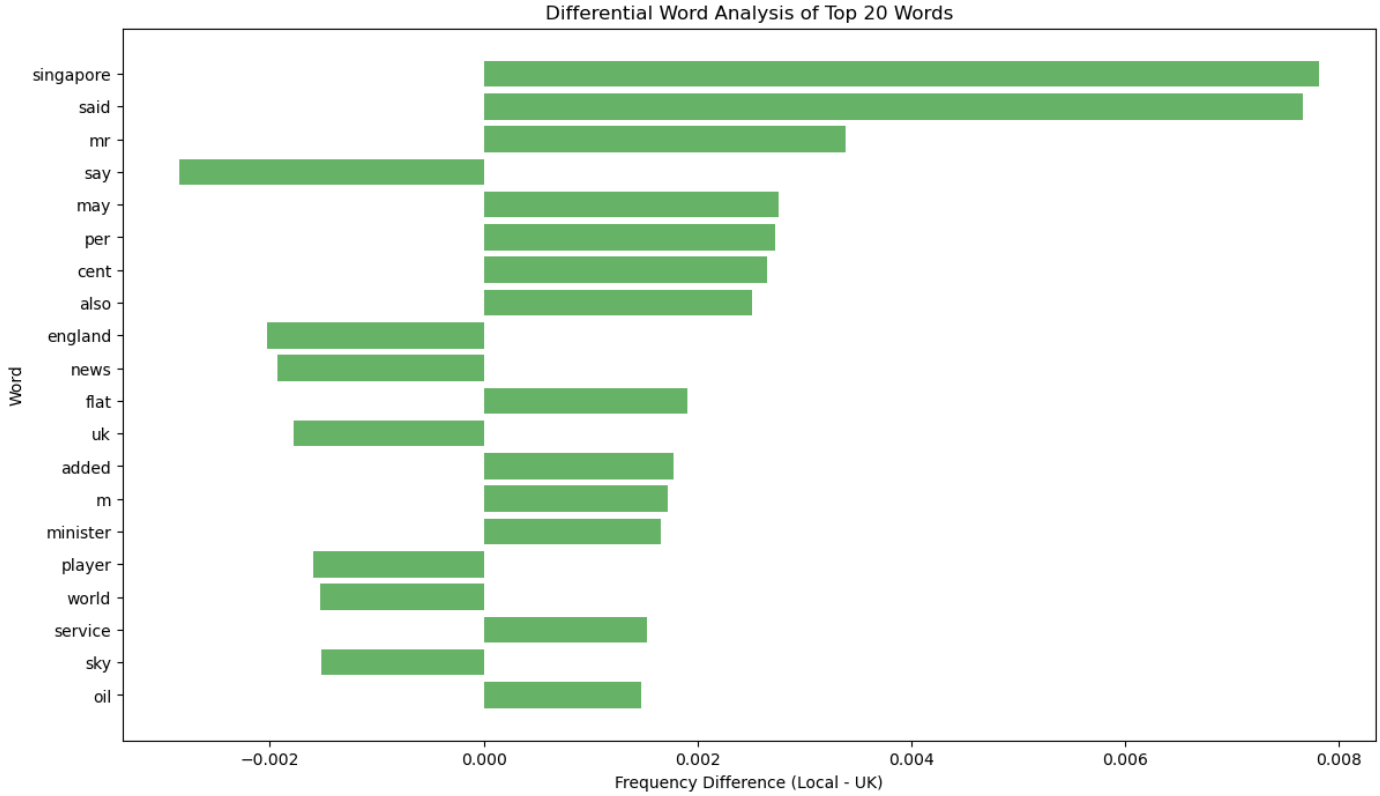
# Calculate the total counts for normalization
total_local = df_local['Count'].sum()
total_uk = df_uk['Count'].sum()

# Normalize the counts
df_merged['Count_local_normalized'] = df_merged['Count_local'] / total_local
df_merged['Count_uk_normalized'] = df_merged['Count_uk'] / total_uk

# Calculate the difference in normalized frequencies
df_merged['Frequency_Difference'] = df_merged['Count_local_normalized'] - df_merged['Cou

# Sort by the absolute value of the frequency difference for better visualization
df_merged_sorted_diff = df_merged.reindex(df_merged['Frequency_Difference'].abs().sort_v

# Plot the differential word analysis
plt.figure(figsize=(14, 8))
plt.barh(df_merged_sorted_diff['Word'], df_merged_sorted_diff['Frequency_Difference'], c
plt.xlabel('Frequency Difference (Local - UK)')
plt.ylabel('Word')
plt.title('Differential Word Analysis of Top 20 Words')
plt.gca().invert_yaxis()
plt.show()
```



6. Summary

6.1 Conclusions

In relation to this project on word analysis between news outlets, the analysis has identified certain words that are frequently found in both Singaporean and UK journalism. However, it is important to note that without a larger dataset, the conclusions drawn from our analysis may not be entirely definitive.

The current dataset, while informative, may be influenced by outliers which could potentially skew the data. These anomalies might result from the limited scope of the sample, affecting the accuracy and reliability of the findings. To mitigate this, a more extensive collection of articles from a wider range of sources and over a longer period would be necessary to ensure a more robust and representative analysis.

Expanding the dataset would allow for a more comprehensive examination of word usage patterns, reducing the impact of outliers and providing a clearer picture of the linguistic tendencies in both Singaporean and UK journalism. This would ultimately enhance the validity of our comparative analysis and provide more conclusive insights into the similarities and differences in word usage between the two regions.

7. References and Resources

7.1 References

[1]The top publishers in Singapore for Q2 2023: https://www.newswhip.com/2023/07/the-top-publishers-in-singapore-for-q2-2023/
[2]Top 100 media outlets in the UK: https://prowly.com/profiles/outlet-lists/top-100-media-outlets-in-the-uk

7.2 Resources

Bird, S., Klein, E., & Loper, E. (2023). Natural Language Toolkit. Retrieved from https://www.nltk.org/